COURSE INTRODUCTION

Big Data in Cloud Computing

Lecturer: Pham Do Kim Chi (Ph.D.)

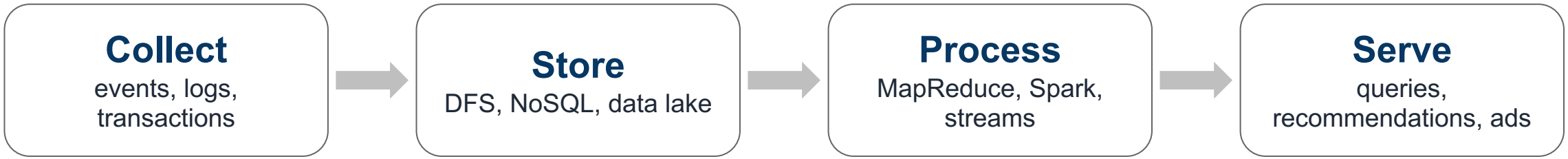
Contents

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The design question

How do you build a system for very large data on someone else's hardware, so that it is affordable, keeps working when machines fail, and grows when the load grows?

Every topic sits on one data axis, and every stage carries the same trade-off: cost, performance, scalability, and operational complexity.



Part 1, Part 8 (streams)

Part 3, 4, 5, 6 (GFS, Dynamo, BigTable, Cassandra, graph)

Part 6 (PaaS), Part 8, Labs

Project

The syllabus map

Part	Cloud thread	Data thread
Part 1		Introduction to Big Data. GFS
Part 2	Basic concepts of cloud computing (NIST definition, deployment and service models, characteristics, enabling technologies)	
Part 3		Data storage models: NoSQL, consistency, DHT, key-value (Dynamo)
Part 4	IaaS model and virtualization (server, storage, network), IaaS providers	Column-family storage (BigTable)
Part 5	Cloud computing simulation system	Document storage (Cassandra)
Part 6	PaaS model and support technologies (file system, programming model, database), PaaS providers	Graph storage models
Part 7		Scalable algorithms: recommendation systems (Amazon item-to-item, Netflix Prize), link analysis (PageRank, spam farms)
Part 8	SaaS model and support technologies (service layer, UI), SaaS providers	Mining data streams. Hadoop mini tutorial
Part 9	Challenges of cloud computing: security, standards, law	Advertising on the web (MMDS ch. 8)
Final	Project presentations	

The program outcomes

G1 Understand and describe the concepts. Cloud computing: definitions, characteristics, deployment models, service models, enabling technologies, and the challenges. Large-scale data: key-value, column-family, and graph storage systems, data stream analysis, and recommendation algorithms.
Competency NT4, program outcome 2.7.

G2 Analyze and choose the service model that fulfills the organizational requirements. Program on a cloud platform and apply new cloud technologies effectively in your work.
Competency KN4, program outcomes 3.1 to 3.3.

G3 Write a technical report for your project in English and present it well.
Competency KN4, program outcomes 6.1 and 6.2.

Outcomes in working form

Statement	Delivers
CLO1	Explain cloud concepts: the NIST model, deployment and service models, enabling technologies, and the main challenges.
CLO2	Choose and justify non-relational storage (key-value, column-family, document, graph) with its consistency trade-offs.
CLO3	Design and implement MapReduce and Spark computations on HDFS-class storage.
CLO4	Apply large-scale analytics algorithms: recommendation, PageRank, stream mining, online advertising.
CLO5	Select a service model and architecture for a given problem, with a cost estimate.
CLO6	Build an end-to-end big data solution on a public cloud.
CLO7	Write a technical report in English, present it, and work in a team.

The final project challenge

Task. Build an end-to-end big data solution deployed either on a public cloud (GCP) or in a cloud simulator (CloudSim):

1. Public big data sets, or:
2. E-commerce behaviour: on the Kaggle "eCommerce Behavior Data from Multi Category Store" dataset (Kechinov, 2020).
3. Fraud detection: large-scale anomaly detection on the PaySim mobile money dataset (Lopez-Rojas et al., 2016).

Groups. [2 to 5] students. Groups formed at S1, proposal due S7, check-in at S10.

Deployment options.

- Public cloud: run the real pipeline on GCP, report measured runtime and billed cost.
- Simulator: implement the pipeline logic locally, model the deployment in CloudSim, report simulated runtime and cost under at least two configurations. Both options are graded with the same rubric. A simulator project must justify its resource model with real price and performance figures.

The final project challenge (cont.)

Deliverables.

- Working code and deployment notes, reproducible on GCP or in the simulator.
- Technical report in English, [8 to 12] pages: problem, architecture, service model choice with cost estimate, storage choice with trade-offs, algorithm, results, lessons.
- 15-minute presentation at S11, all members speak.

Rubric.

- Technical solution and results: 40%
- Architecture and cost argument (CLO5): 20%
- Report quality in English: 20%
- Presentation and teamwork: 20% Individual marks may be adjusted by peer evaluation.

Grading

Component	Weight	Form
Lab assignments (5-6, equal weight)	20%	Individual
Final project challenge	30%	Group (2-5)
Weekly opening quiz	up to +5% bonus to the assignments.	Individual
Final exam	50%	Individual, written, in English

Materials

Readings.

- Main text: Mining of Massive Datasets, 3rd ed. (Leskovec, Rajaraman, Ullman, 2020). PDF and videos at mmds.org. Selected chapters.
- Cloud reference: Buyya, Broberg, Goscinski. Cloud Computing: Principles and Paradigms, Wiley, 2011. Selected chapters. NIST SP 800-145.
- Systems papers, read from the source: GFS (2003), Dynamo (2007), Bigtable (2006), Cassandra (2010).
- Spark and labs: Learning Spark, 2nd ed. (Damji et al., 2020), free from Databricks. Google Cloud documentation.
- Algorithm weeks adapt Stanford CS246 material with video links for self-study.

Tools and accounts.

- Google Cloud Platform: [free tier or education credits, how to claim].
- Apache Spark (PySpark), Hadoop HDFS, one NoSQL store of your choice for the project.
- LMS for slides, readings, submissions, and the quiz.
- [Communication channel] for questions.

Reading load. About one paper every two weeks, assigned on the homework slide of each deck.

Course policies

- **Language**: the course, the labs, the report and the exam are in English.
- **Attendance**: all sessions are expected. Maximum absence is 6 hours per semester. Excessive absences carry a grade penalty. Excused absences only for illness or family emergency, requested in person and in writing.
- **Labs are individual work**. Discussion is encouraged, copied code, or write-ups receive zero for all parties.
- **Project work is shared inside the group**. Cite every external source, dataset, and library.
- **AI tools:** students may use AI to support labs and the report. Every submission must include a short AI-use statement marking, part by part, what was AI-assisted and what was done independently. **The instructor may ask the student to explain any part of the submission.** A student who can explain and defend their work passes; one who cannot is treated as if they did not do the work themselves, regardless of the AI-use statement.
- **Late submissions**: -5% per day, up to 10 days.

References

Leskovec, Jure, Anand Rajaraman, and Jeffrey D. Ullman. Mining of Massive Datasets. 3rd ed., Cambridge UP, 2020, mmds.org.

Mell, Peter, and Timothy Grance. The NIST Definition of Cloud Computing. NIST SP 800-145, 2011.

Ghemawat, Sanjay, Howard Gobioff, and Shun-Tak Leung. The Google File System. SOSP, 2003.

DeCandia, Giuseppe, et al. Dynamo: Amazon's Highly Available Key-value Store. SOSP, 2007.

Chang, Fay, et al. Bigtable: A Distributed Storage System for Structured Data. OSDI, 2006.

Lakshman, Avinash, and Prashant Malik. Cassandra: A Decentralized Structured Storage System. ACM SIGOPS OSR, 2010.

Damji, Jules S., et al. Learning Spark. 2nd ed., O'Reilly, 2020.

Kechinov, Mikhail. eCommerce Behavior Data from Multi Category Store. Kaggle, 2020.

Lopez-Rojas, Edgar, Ahmad Elmir, and Stefan Axelsson. PaySim: A Financial Mobile Money Simulator for Fraud Detection. EMSS, 2016.

Stanford CS246: Mining Massive Data Sets. Course materials, http://www.mmds.org.